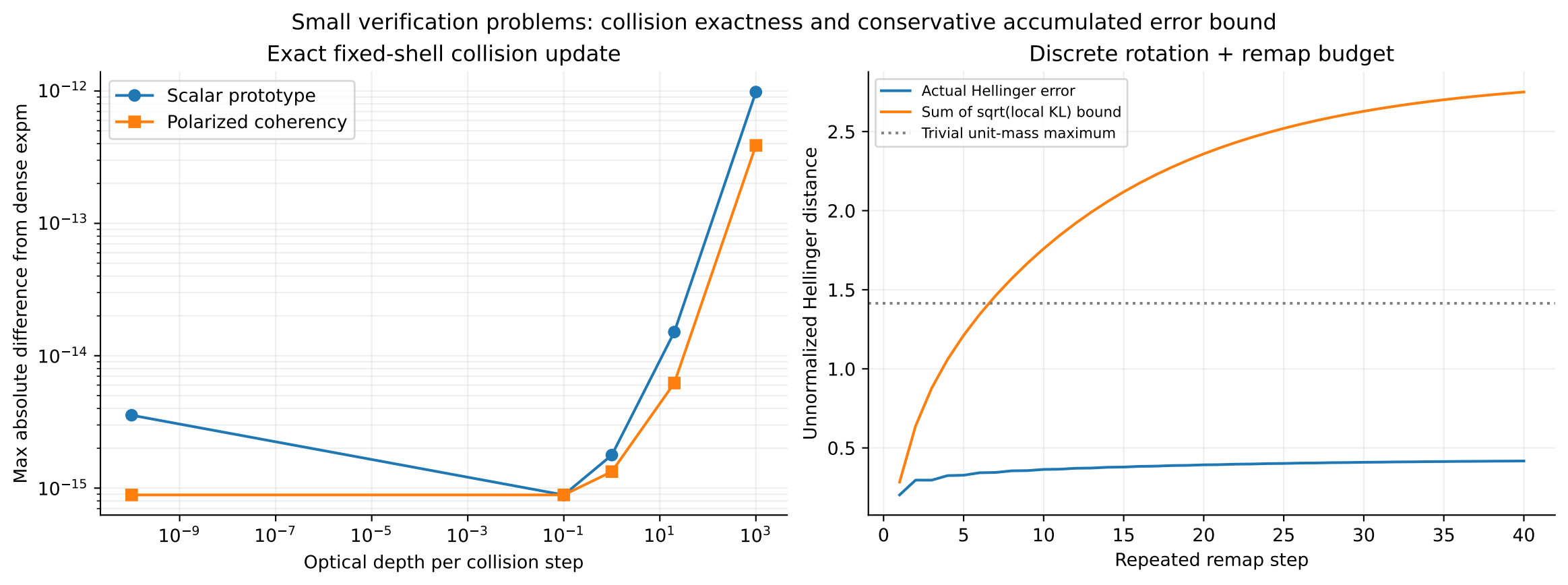
Small verification problems: collision exactness and conservative accumulated error bound
Exact fixed-shell collision update
Scalar prototype
Polarized coherency
Max absolute difference from dense expm
$10^{-12}$
$10^{-13}$
$10^{-14}$
$10^{-15}$
$10^{-9}$
$10^{-7}$
$10^{-5}$
$10^{-3}$
$10^{-1}$
$10^{1}$
$10^{3}$
Optical depth per collision step
Discrete rotation + remap budget
Actual Hellinger error
Sum of sqrt(local KL) bound
Trivial unit-mass maximum
Unnormalized Hellinger distance
0.5
1.0
1.5
2.0
2.5
0
5
10
15
20
25
30
35
40
Repeated remap step